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1. Compare Permissions on Regular Files and Directories

Default Permissions	User	Group	Others
File	read,write	read	read
Directory	read,write, execute	write, execute	execute

```
(kali㉿kali)-[~]
$ touch sample.txt

(kali㉿kali)-[~]
$ ls -l sample.txt
-rw-rw-r-- 1 kali kali 0 Aug 22 21:40 sample.txt

(kali㉿kali)-[~]
$ mkdir sample_dir

(kali㉿kali)-[~]
$ ls -l sample_dir
total 0

(kali㉿kali)-[~]
$ ls -ld sample_dir
drwxrwxr-x 2 kali kali 4096 Aug 22 21:40 sample_dir
```

2. Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

- Read: cat sample.txt
- Write: echo “writing” > sample.txt
- Change permission: chmod 000 sample.txt

* Execute permission allowed only for executable files for sample.txt it is invalid

Permission	Regular File (sample.txt) – Test	Observation
400 (r--)	Try to read and modify the file	• read - allowed • write - when tried to modify shows ‘readonly’
200 (-w-)	Try to read and modify the file	• read - not allowed • write - possible using echo >sample.txt and using vi not able to write
100 (--x)	Try to read and execute the file	• read - denied • write – denied (allowed only for executable files)

500 (r-x)	Try to read and modify the file	• read- allowed • write- Permission denied • execute- allowed(but invalid)
600 (rw-)	Try to read and modify the file	• read- allowed • write- allowed • execute- not allowed
700 (rwx)	Try to read, modify, and execute the file	• read and write allowed

➤ DIRECTORY

- r = list
- w = create/delete/rename entries
- x = enter/traverse the directory

Permission	Directory (sample_dir) – Test	Observation
400 (r--)	Try to list and enter the directory	• dr----- 2 sa sa 4096 Aug 22 10:23 sample_dir • list: ls: cannot access 'sample_dir/file1.txt': Permission denied file1.txt • enter: denied
200 (-w-)	Try to list, enter, and create a file	• list: denied • enter: denied • create: denied
100 (--x)	Try to list and enter the directory	• list: denied • enter: denied
500 (r-x)	Try to list and enter the directory	• list: denied • enter: allowed and can access the files
600 (rw-)	Try to list, enter, and create a file	• list: denied • enter: denied • create: denied <pre>ls -l sample_dir ls: cannot access 'sample_dir/file1.txt': Permission denied total 0 -???????? ? ? ? ? ? file1.txt</pre>
700 (rwx)	Try to list, enter, create, and delete files	All allowed

3. Questions

- What does read (r) permission allow you to do with a regular file?

Allows read/view content of the file.

- What does read (r) permission allow you to do with a directory?

List the names of the files and directories.

- What does write (w) permission mean for a regular file?

Modify, overwrite, or write data to the file.

- What does write (w) permission mean for a directory?

Create, delete, and rename files or directories inside it.

- What does execute (x) permission mean for a regular file?

Allows the file to be executed as a program/script, provided it is a valid executable or script.

- Why is execute (x) permission important for accessing a directory?

Enter/traverse the directory and access items inside it.

- Which permission combination allows the owner to fully access a regular file

rwX (700)

- Which permission combination allows the owner to fully access a directory??

rwX (700)

- Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

File contains data and directory contains entries like files, directories. Therefore, the permissions have different meanings.

- File: **r** means reading contents, **w** means modifying contents and **x** means execute/run scripts or files.
- Directory: **r** means listing entries, **w** means creating/deleting entries, and **x** means entering or traversing the directory.